NBA Team Performance and Three-Point Shooting Efficiency

A Fixed Effects Analysis of the Modern NBA (2014–2024)

Drew Combs

Sports Analytics Research Project

Temple University

Department of Economics

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Executive Summary

This study examines the relationship between three-point shooting efficiency and team success in the modern NBA. Using team-level data from the 2013–14 through 2023–24 NBA seasons, the analysis evaluates whether teams that outperform the league average in weighted three-point efficiency achieve more wins after controlling for additional offensive and defensive variables. To isolate team performance relative to league-wide trends, a fixed effects regression model was implemented using season-demeaned variables. The model incorporated multiple offensive and defensive controls, including turnovers, rebounds, assists, steals, blocks, and personal fouls. The results indicate that weighted three-point efficiency is a statistically significant positive predictor of team success. Defensive metrics such as steals, blocks, and defensive rebounds also contributed positively to winning, while turnovers and personal fouls negatively impacted team performance. Overall, the model explains approximately 67% of the variation in NBA team wins, reinforcing the growing importance of shooting efficiency, possession management, and defensive disruption in the modern NBA.

Key Findings

- Weighted three-point efficiency significantly predicts NBA team wins.
- Turnovers negatively impact team success.
- Defensive rebounds, steals, and blocks remain highly valuable.
- Assists and offensive rebounds were not statistically significant after controlling for efficiency metrics.
- The model explains approximately 67% of variation in team wins.

Introduction

Over the last decade, the NBA has experienced a major strategic transformation driven by the increasing importance of three-point shooting and offensive spacing. Teams have shifted away from midrange-heavy offenses toward systems emphasizing efficient shot selection, floor spacing, and high-volume perimeter shooting. As a result, three-point attempts and leaguewide shooting efficiency have both increased dramatically.

This project examines whether outperforming the league average in three-point shooting efficiency translates into greater team success. Rather than focusing solely on raw three-point volume, this study evaluates weighted three-point efficiency, which combines both shooting percentage and shot volume. The analysis also controls for several offensive and defensive variables including turnovers, rebounds, assists, steals, blocks, and personal fouls.

The goal of the study is to determine how strongly modern shooting efficiency contributes to winning while accounting for team-specific differences and league-wide seasonal trends.

Three-Point Percentage vs Wins (2014–2024)

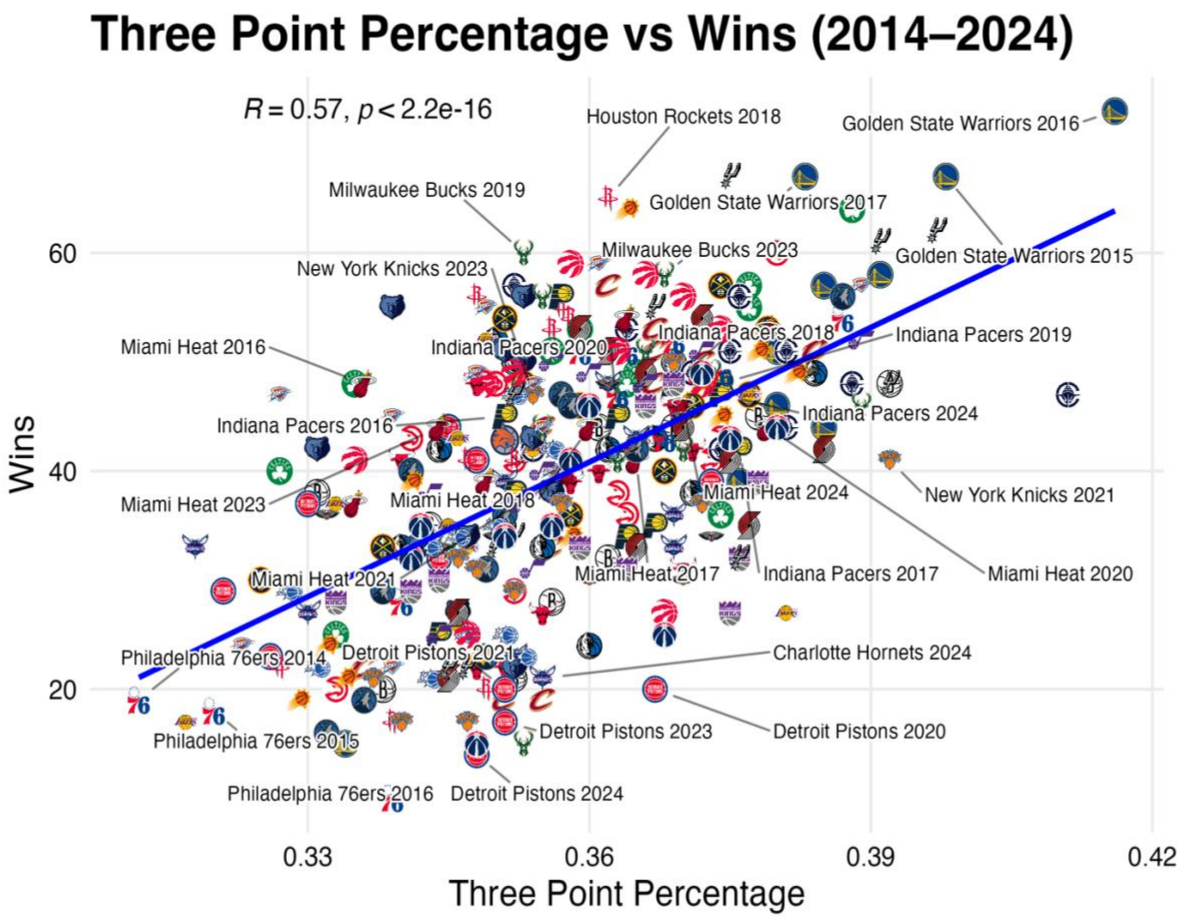

Figure 1. Relationship between three-point shooting percentage and NBA team wins from 2014–2024. Teams with higher three-point efficiency generally achieved more regular season wins.

Related Research and NBA Context

Previous research has documented the NBA's long-term transition toward perimeter-oriented offenses. A 2023 study titled *Long-Term Trends in Shooting Performance in the NBA* analyzed league-wide shooting trends from 1979–80 through 2018–19 and found a dramatic increase in three-point attempts and efficiency over time. The study highlighted the league's movement toward spacing-oriented offenses and away from traditional midrange scoring.

This project focuses specifically on the modern NBA era from 2013–14 through 2023–24, which represents the height of the league's three-point explosion. However, rather than examining historical trends over multiple eras, this analysis focuses on differences between teams within each season. Since nearly every team now emphasizes three-point shooting, the study investigates whether teams that shoot more efficiently relative to their competition gain a meaningful competitive advantage in wins.

The modern NBA places a premium on possession efficiency, shot quality, and spacing. Because of this, shooting efficiency may provide more explanatory power than raw shot volume alone.

Research Question

Does outperforming the league average in weighted three-point shooting efficiency lead to more wins in the NBA?

Hypothesis

The primary variable of interest is weighted three-point efficiency:

$$Weighted\ 3P\ Efficiency = 3P\% \times 3PA$$

This variable combines both three-point accuracy and shot volume.

Because modern NBA offenses prioritize efficient three-point shooting and floor spacing, teams that exceed league-average three-point efficiency are expected to win more games.

Therefore:

- **Null Hypothesis (H_0):** Weighted three-point efficiency has no effect on wins.
- **Alternative Hypothesis (H_1):** Weighted three-point efficiency positively affects wins.

Data and Methodology

The dataset contains team-level NBA statistics from Basketball Reference spanning the 2013–14 through 2023–24 NBA seasons. Each observation represents a single NBA team season, creating a panel dataset covering multiple franchises across multiple years.

The analysis focuses on the relationship between weighted three-point shooting efficiency and team success while controlling for additional offensive and defensive performance metrics. Key variables include three-point percentage, three-point attempts, two-point efficiency, free throw efficiency, turnovers, rebounds, assists, steals, blocks, and personal fouls.

To reduce the impact of league-wide strategic changes over time, all efficiency and control variables were demeaned by season. This approach isolates team performance relative to the league average within each individual NBA season rather than comparing teams across fundamentally different offensive eras.

A fixed effects regression framework was implemented to account for unobserved franchiselevel characteristics such as organizational structure, player development, coaching stability, and roster construction. Team fixed effects help isolate within-team variation over time while controlling for persistent organizational differences between franchises.

The primary regression model is specified as:

$$Wins\_it = \beta_0 + \beta_1 W3P\_it + \beta_2 TwoP\_it + \beta_3 FT\_it + \beta_4 TOV\_it + \beta_5 ORB\_it + \beta_6 AST\_it + \beta_7 STL\_it + \beta_8 BLK\_it + \beta_9 DRB\_it + \beta_{10} PF\_it + Team\_i + \varepsilon\_it$$

where Wins_it represents team wins for team i in season t, and Team_i captures franchise fixed effects.

Model Construction and Statistical Considerations

Several steps were taken to improve the interpretability of the model and reduce omitted variable bias.

The initial model relied heavily on raw volume statistics such as total three-pointers made and attempted. However, variance inflation factor (VIF) testing revealed multicollinearity concerns, particularly involving raw shooting volume variables.

To address this issue:

- weighted shooting efficiency variables replaced raw volume variables
- variables were demeaned by season
- additional defensive controls were added

These adjustments helped isolate team-level performance differences from league-wide strategic trends.

However, potential endogeneity concerns remain. Winning teams may generate more efficient shot opportunities, and unobserved variables such as injuries, coaching adjustments, roster talent, or schedule strength may still influence results.

Future research could incorporate:

- travel fatigue variables
- opponent defensive quality
- player tracking data
- injury-adjusted lineup metrics
- instrumental variable approaches

Regression Results

The fixed effects regression model produced statistically significant results overall.

Model Fit

- R-squared: 0.670
- Adjusted R-squared: 0.625
- F-statistic: 14.83
- p-value: < 0.001

The model explains approximately 67% of the variation in NBA team wins across the sample.

Significant Positive Predictors of Wins

- Weighted Three-Point Efficiency
- Two-Point Efficiency
- Free Throw Efficiency
- Steals
- Blocks
- Defensive Rebounds

Significant Negative Predictors of Wins

- Turnovers
- Personal Fouls

Statistically Insignificant Variables

- Offensive Rebounds
- Assists

The results indicate that shooting efficiency, possession management, and defensive disruption strongly contribute to team success in the modern NBA.

Table 1. Fixed Effects Regression Results

Variable	**Coefficient**	**p-value**	**Effect on Wins**
Weighted 3P Efficiency	+0.036	< 0.001	Positive
Two-Point Efficiency	+0.041	< 0.001	Positive
Free Throw Efficiency	+0.028	< 0.001	Positive
Turnovers	-0.045	< 0.001	Negative
Offensive Rebounds	+0.006	Not Significant	Neutral
Assists	+0.004	Not Significant	Neutral
Steals	+0.049	< 0.001	Positive
Blocks	+0.017	< 0.05	Positive
Defensive Rebounds	+0.029	< 0.01	Positive
Personal Fouls	-0.015	< 0.05	Negative

Note: The model includes team fixed effects and season-demeaned variables to control for league-wide strategic trends over time.

Visual Relationship Across Seasons (2014-2024)

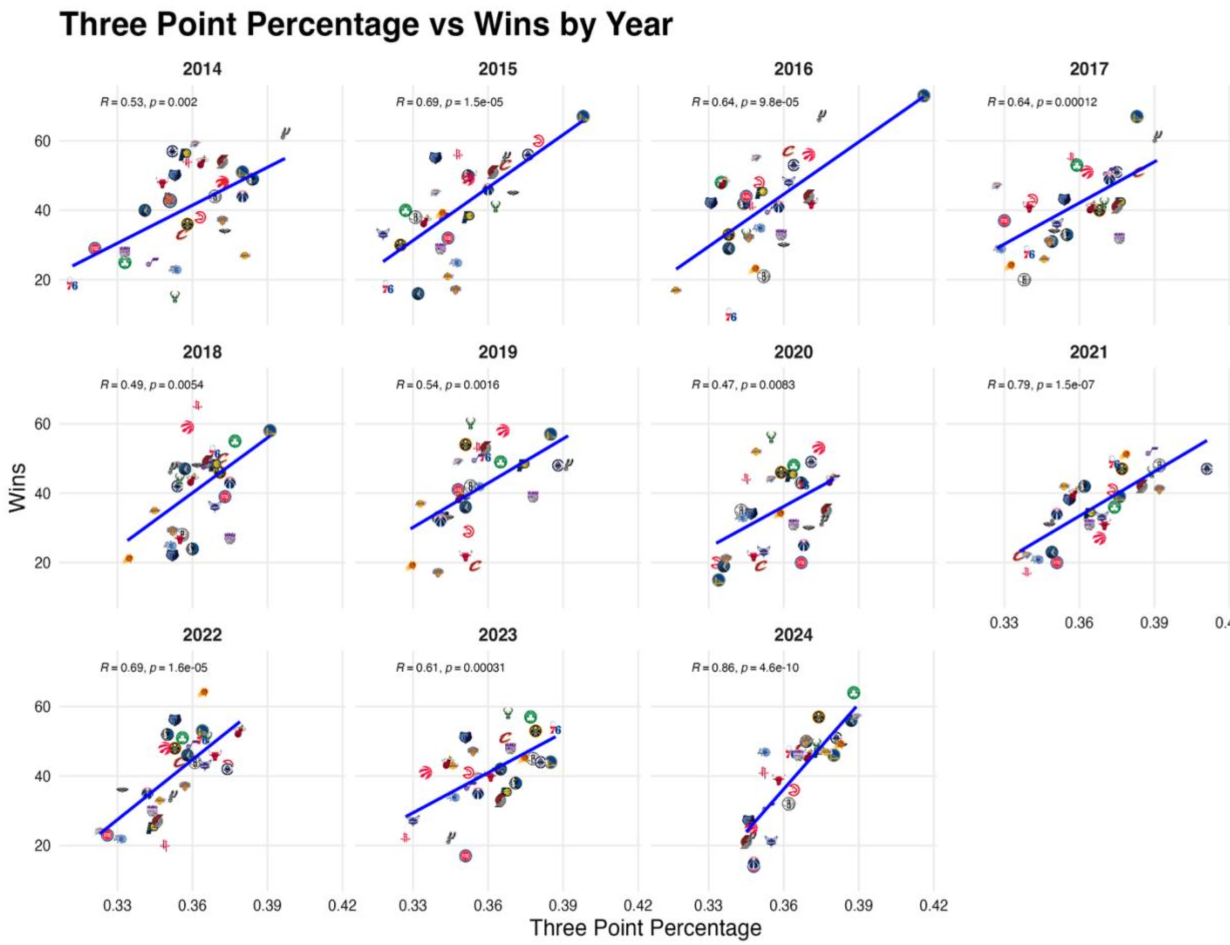

Figure 2. Relationship between three-point percentage and team wins across individual NBA seasons from 2014– 2024.

Basketball Interpretation

The findings strongly support the importance of efficient perimeter shooting in the modern NBA. Teams that outperform the league average in weighted three-point efficiency consistently win more games, even after controlling for other offensive and defensive factors.

The significance of steals, blocks, and defensive rebounds suggests that defense remains highly valuable despite the league's offensive evolution. Teams that generate extra possessions and successfully finish defensive possessions gain meaningful advantages over opponents.

Turnovers were among the strongest negative predictors of wins, reinforcing the increasing value of possessions in modern NBA offenses.

One of the more interesting findings is that assists and offensive rebounds were not statistically significant after controlling for efficiency variables. This may suggest that ball movement primarily matters because it creates efficient scoring opportunities rather than because assists themselves independently drive winning. Similarly, many modern teams sacrifice offensive rebounding opportunities in favor of transition defense and spacing.

The team fixed effects also suggest that some organizations consistently outperform expectations relative to league averages, potentially reflecting advantages in coaching, player development, organizational structure, or roster construction.

Research Limitations and Future Directions

Although the model explains a substantial portion of variation in NBA team wins, several limitations remain.

First, the analysis relies on team-level season data rather than player-level or lineup-level information. While this framework is effective for identifying broad organizational trends, it cannot fully capture lineup interactions, rotational adjustments, or situational in-game dynamics that may influence team performance.

Second, the model cannot completely account for contextual variables such as injuries, roster changes, coaching adjustments, schedule strength, or opponent quality. Many of these factors fluctuate throughout a season and may affect both shooting efficiency and overall team success.

Third, the analysis identifies statistical relationships between variables and winning but does not establish definitive causality. Teams that win more games may naturally generate more efficient offensive opportunities, creating potential endogeneity concerns within the regression framework.

Future research could strengthen the analysis by incorporating player tracking data, lineup adjusted metrics, shot quality models, travel fatigue variables, and machine learning approaches capable of capturing more complex nonlinear relationships within NBA team performance.

Conclusion

This study demonstrates that shooting efficiency, particularly from three-point range, plays a major role in NBA team success during the modern spacing era. Teams that outperform the league average in weighted three-point efficiency consistently win more games, even after controlling for additional offensive and defensive factors. The results also reinforce the continued importance of defense, possession control, and efficiency-based basketball strategy. Defensive rebounds, steals, and blocks significantly contributed to winning, while turnovers and fouls negatively impacted team success. Overall, the findings reflect the strategic evolution of the modern NBA toward efficient shot selection, spacing, and possession optimization. The model's strong statistical fit and interpretable basketball logic suggest that weighted shooting efficiency provides meaningful insight into team performance across the modern NBA landscape.

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Repository and Reproducibility

GitHub Repository:

github.com/drewcombs/nba-analytics-project

The full codebase, cleaned datasets, regression models, and visualizations used in this study are publicly available through the linked GitHub repository. The repository includes the R scripts used for data cleaning, fixed effects regression modeling, diagnostic testing, and data visualization.